

07/11/14(E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)
Semester: July 2014 - November 2014

Max. Marks:100

Duration:3hrs

End Semester Exam

Class: SE

Semester: III

Name of the Course: Data Structures and Algorithm Analysis

Branch: IT

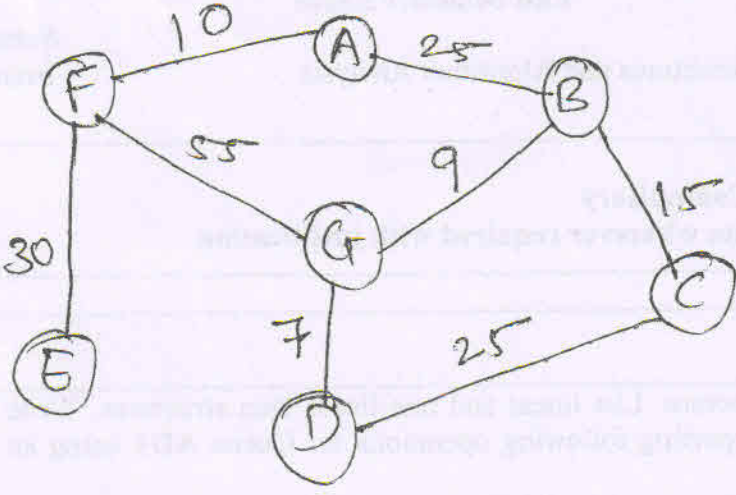
Course Code: UITC302

Instructions:

(1) All Questions are Compulsory

(2) Assume suitable data wherever required with justification

Question No.		Max. Marks
Q1(a)	Define data structure. List linear and non-linear data structures. Write C functions supporting following operations for Queue ADT using an array. i) create, ii) enqueue, iii) dequeue, iv) isempty OR What is Priority Queue? Write an algorithm to implement descending priority queue using heap. Specify time complexity for following operations on it i) insert, ii) deleteMax, iii) search, iv) display	10
Q1(b)	What is Asymptotic analysis? Explain asymptotic notations for Big Oh, Big Omega and Big Theta.	10
Q2(a)	Write an algorithm to implement Quick sort. Show it's working with the help of following dataset - 14, 10, 1, 20, 17, 24, 18, 12, 15, 11 OR Write a C program to implement Merge sort.	10
Q2(b)	Compare insertion, quick, merge and radix sort algorithms. Suggest best suitable sorting algorithm for adding new student names in alphabetically sorted exiting list.	10
Q3(a)	Write an algorithm to convert Infix expression to equivalent Postfix expression. Show the use of Stack data structure in the above algorithm with the help of an example. OR What is Recursion? Explain the use of Stack in recursion. Give advantages and disadvantages of recursion.	10
Q3(b)	What are Static and Dynamic implementations? Write C functions to implement following operations on Doubly Linked List i)insert before, ii) delete after, iii) delete an element, iv) display	10
Q4(a)	What is the drawback of Binary Search Tree? Construct solution tree to overcome this drawback using following dataset - Order: 14, 15, 16, 13, 12, 11, 10, 1, 2, 3, 4 OR What is B-Tree? Construct 5-way B-tree for following dataset – order:1 12 8 2 25 5 14 28 17 7 52 16 48 68 3 26 29 53 55 45	10
Q4(b)	What is a Tree data structure? Justify its efficiency as a data structure. Write C functions for following operations on Binary Search Tree i) insert, ii) delete, iii) traverse in Pre-order fashion	10

Q5(a)	Represent following graph using the data structures used for representing graph. Draw minimum spanning tree and calculate its cost using Kruskal's algorithm for the same. 	10
Q5(b)	What are the graph traversal techniques? Apply DFS on graph given in Q5(a) and show the use of appropriate data structure.	10